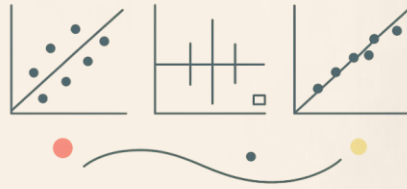


Interactive Diagnostics for Linear and Logistic Regression Models



Design, Implementation and Validation of the `diag_lm` Framework

Rezaul Hoque · Sarah Al-Talal · Thi Hong Cam Nguyen

University of Duisburg-Essen — Chair of Econometrics

Presentation Overview

01 · Motivation — Why Model Diagnostics Matter

02 · The diag_lm Framework

03 · Different Models Need Different Diagnostics

04 · Validation with Known Ground Truth

05 · Linear Regression Validation

06 · Time-Series Validation

07 · Logistic Regression Validation

08 · From Framework to Interactive Application

09 · From Statistical Tests to Visual Diagnostics

10 · Statistics First — AI Interpretation Second

11 · Real-Data Case Study: Hourly Wages

12 · Diagnostics Guide Model Improvement

13 · What Did We Achieve?

14 · Thank You & Questions

Why Model Diagnostics Matter



01

Assumption Violations

Heteroskedasticity, serial correlation, non-normality, and multicollinearity can undermine statistical inference.



02

Different Models

Linear, logistic, time-series, and panel models each require distinct diagnostic procedures.



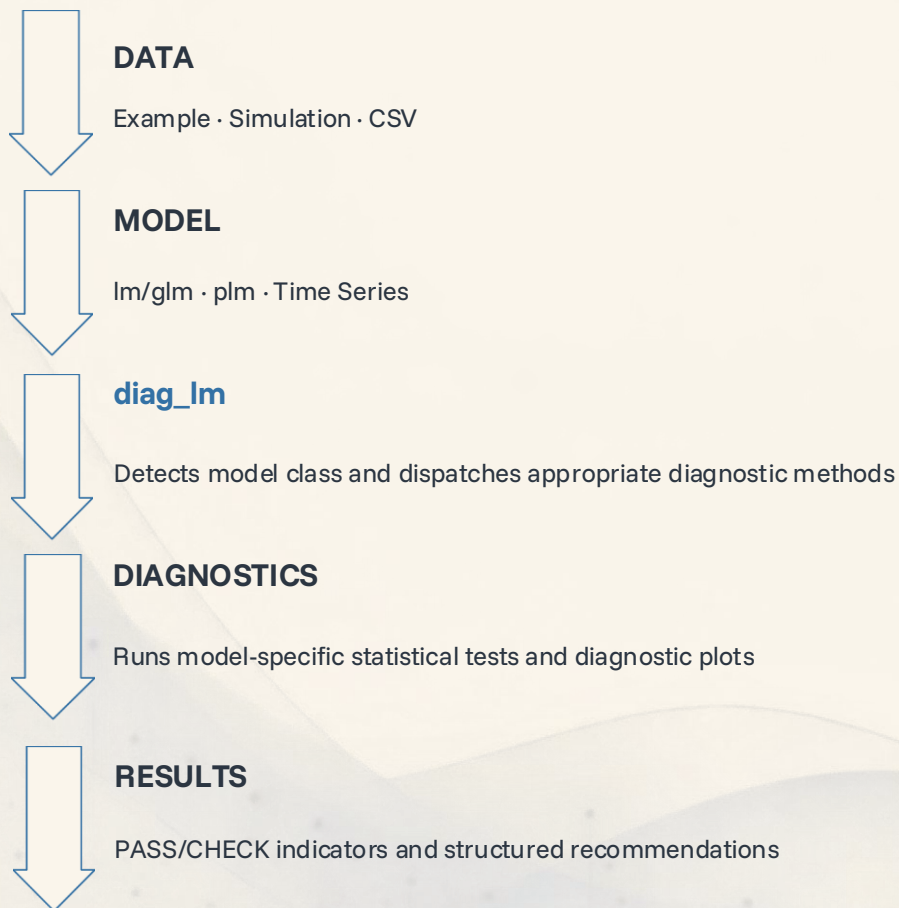
03

Manual Workflow

Running and interpreting separate tests repeatedly is time-consuming and error-prone.

Our Goal: Create one reusable, interactive, and model-aware framework for regression diagnostics.

The diag_lm Framework



S3 Architecture

Automatic model-specific
method dispatch

Modular Computation

Diagnostics are reusable
and independently
testable

Consistent Interface

One workflow across multiple model families

How it connects to the app: The Shiny app provides the interface; diag_lm acts as the statistical engine. User selections trigger the framework, which returns structured results.

One diagnostic engine powers multiple model-specific workflows within the same interactive application.

Different Models Need Different Diagnostics

01 LINEAR

Breusch-Pagan -
Heteroskedasticity

Durbin-Watson - Serial
correlation

Shapiro-Wilk - Residual
normality

VIF - Multicollinearity

02 LOGISTIC

Hosmer-Lemeshow -
Goodness of fit

ROC/AUC - Discrimination
performance

Calibration - Predicted vs.
observed

VIF - Multicollinearity

03 TIME SERIES

ADF - Stationarity / unit roots

Ljung-Box - Serial
dependence

ACF/PACF - Autocorrelation
structure

Normality - Residual
distribution

04 PANEL

Panel LM - Pooled vs. panel
structure

Hausman - Fixed vs. random
effects

Panel Breusch-Godfrey -
Serial correlation

Time-effects test - Time-
specific effects

Why Time Series and Panel Models Need Separate Diagnostics

Their data structures introduce dependencies that standard cross-sectional diagnostics do not fully address.

TIME SERIES

Observations are ordered over time — errors and values may depend on past behavior.

Serial Correlation

Are residuals correlated across time?

Test: Ljung-Box / Breusch-Godfrey

Stationarity

Are statistical properties stable over time?

Test: ADF test

Dependence Structure

How does correlation behave across lags?

Test: ACF / PACF

PANEL DATA

Entities tracked over time — creating cross-sectional and temporal dependence.

Panel Structure

Is panel specification preferable to pooled OLS?

Test: Panel LM

Fixed vs. Random Effects

Which panel specification is appropriate?

Test: Hausman test

Within-Entity Correlation

Are errors correlated over time within entities?

Test: Panel Breusch-Godfrey



Different data structures create different statistical problems - therefore diag_lm follows model-specific diagnostic paths.

Validation with Known Ground Truth

Validation Process

01

Generate Baseline

Clean data, no violations

03

Diagnose with diag_lm

Run full diagnostic suite

02

Inject Violation

Introduce known assumption breach

04

Verify Against Truth

Compare results to known ground truth

Violations Tested

01 · Heteroskedasticity

02 · Multicollinearity

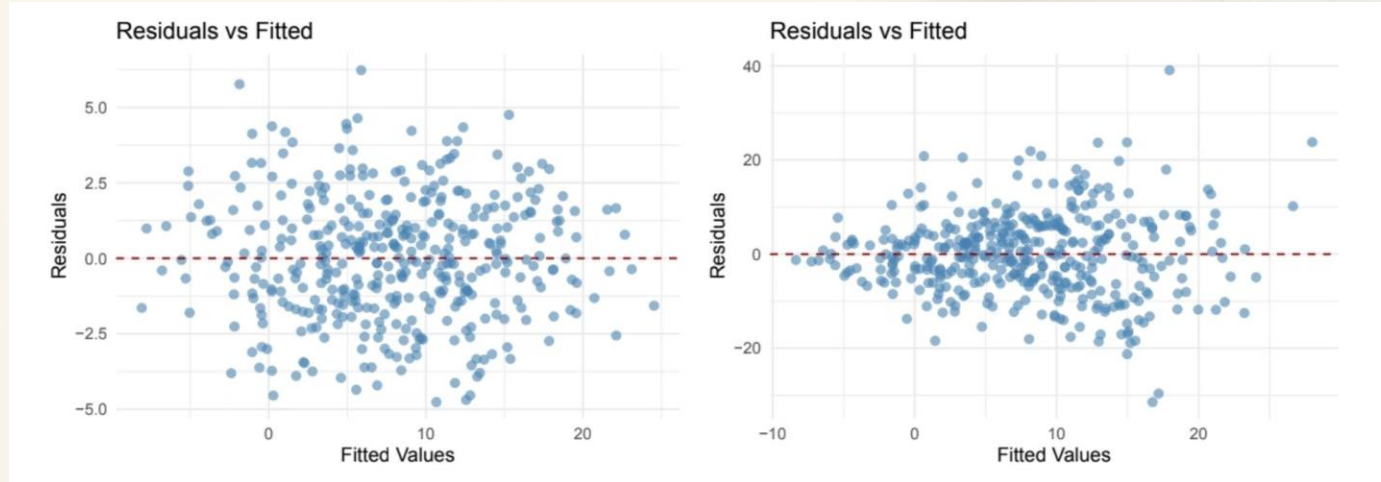
03 · AR(1) Serial Correlation

04 · Unit Roots

05 · Panel Effects

Linear Regression Validation

Injected violations produce the expected diagnostic response.



BASELINE

BP $p = 0.486$

Random scatter → no heteroskedasticity detected

HETEROSKEDASTICITY INJECTED

BP $p \approx 0$

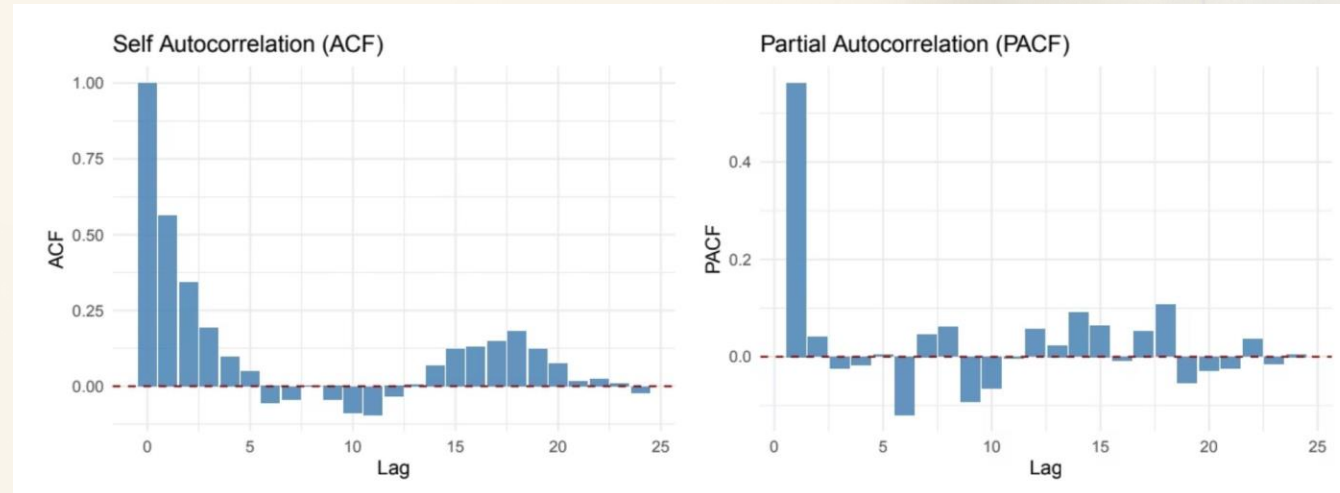
Funnel-shaped spread → violation detected

Additional validation · Multicollinearity injected → Max VIF = 15.68

The diagnostics respond specifically to the violations introduced in simulation.

Time-Series Validation: Detecting Serial Correlation

AR(1) dependence was deliberately introduced to test whether diag_lm detects it.



ACF

Slowly decaying autocorrelation → persistent serial dependence

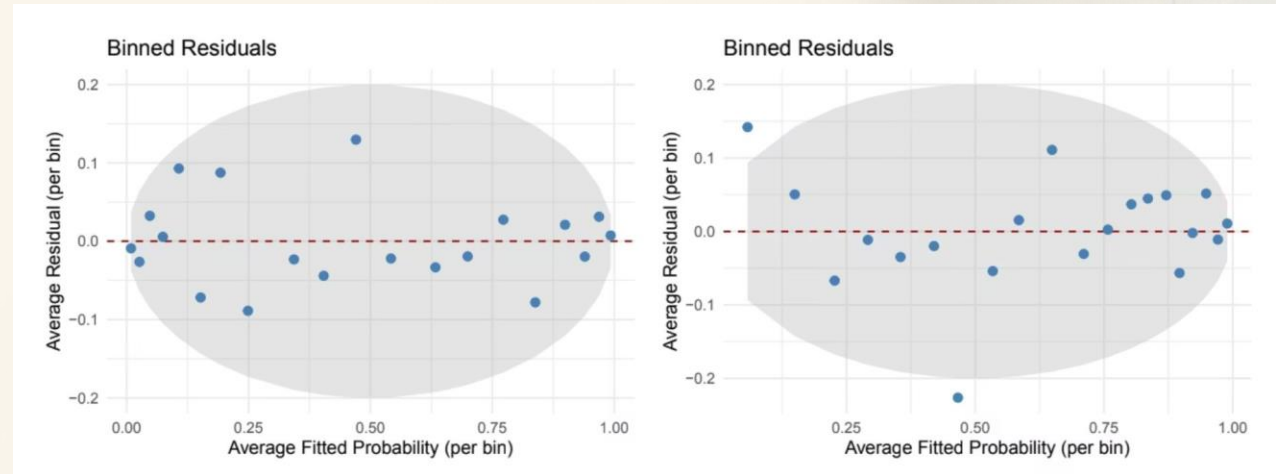
PACF

Strong first lag → pattern consistent with a low-order AR process

Breusch–Godfrey: $p \approx 0$ · Ljung–Box: $p \approx 0$ · Both tests reject the null of no serial correlation.

ADF: Stationary AR(1) $p = 0.010$ · Random Walk $p = 0.073$

Logistic Regression Validation: Detecting Misspecification



BASELINE

Residuals scatter randomly within the reference band → no systematic misspecification

OMITTED NONLINEARITY

Systematic curvature appears → model misspecification detected

Hosmer–Lemeshow: p-value collapses when nonlinearity is omitted · Formal goodness-of-fit confirms the pattern in binned residuals.

Multicollinearity injected → Max VIF = 18.60 · Baseline AUC = 0.894

From Framework to Interactive Application

The Shiny interface turns the diagnostic framework into a guided workflow for data setup, model specification and analysis.

RegressionDiagnostics

1. Setup & Specifications

Data Setup

Model Family: OLS (Linear Regression)

☐ Simulate data

☒ Choose available data

☐ Upload a CSV

Choose Available Dataset: Wage Data (wage.csv)

✓ System file ready: wage.csv

Modify Variable Types

Select variable to edit: person

New data type: Factor

2. Diagnostics

3. Transformation

Data Preview & Statistical Properties

System Metadata & Structural Profile

Data Structure Class: Other Data

Observations Count (N): 8328

Total Variables Shown: 11

Raw Data Grid

person	east	age	minijob	experience	univ	apprent	novoc	female	Limited
1	0	56	17.26	0	18.12	1	0	0	1
2	0	56	16.38	0	20.88	0	1	0	1
3	0	27	8.15	0	0.75	1	0	0	0
4	1	38	43.15	0	12.88	1	0	0	1
5	0	27	12.79	0	4.79	0	0	1	0
6	0	37	35.29	0	15.25	1	0	0	0
7	0	55	15.54	0	27.92	0	1	0	1
8	1	49	27.28	0	18.71	1	0	0	1
9	1	31	20.14	0	4.25	1	0	0	1
10	0	39	15.54	0	3.42	1	0	0	1
11	0	52	25.44	0	28.58	1	0	0	1
12	1	35	34.52	0	5.71	1	0	0	0

Data Column Profiles

No	Variable	Stats / Values	Freqs (% of Valid)	Missing
1	person [integer]	Mean (sd) : 4186.7 (2417.8) min <= med <= max: 1.8 <= 4184.5 <= 8373.8 IQR (CV) : 4187.5 (8.8)	8328 distinct values	0 (0.0%)
2	east [integer]	Mean (sd) : 0.2 (0.4) min <= med <= max: 0.0 <= 0.0 <= 1.0 IQR (CV) : 0.0 (1.9)	2 distinct values	0 (0.0%)
3	age [integer]	Mean (sd) : 44.1 (12.8) min <= med <= max: 18.0 <= 45.0 <= 64.8 IQR (CV) : 19.8 (8.3)	47 distinct values	0 (0.0%)

1 — SETUP

Choose example data, simulate data or upload a CSV

2 — SPECIFY

Select the model family and configure variable types

3 — PREVIEW

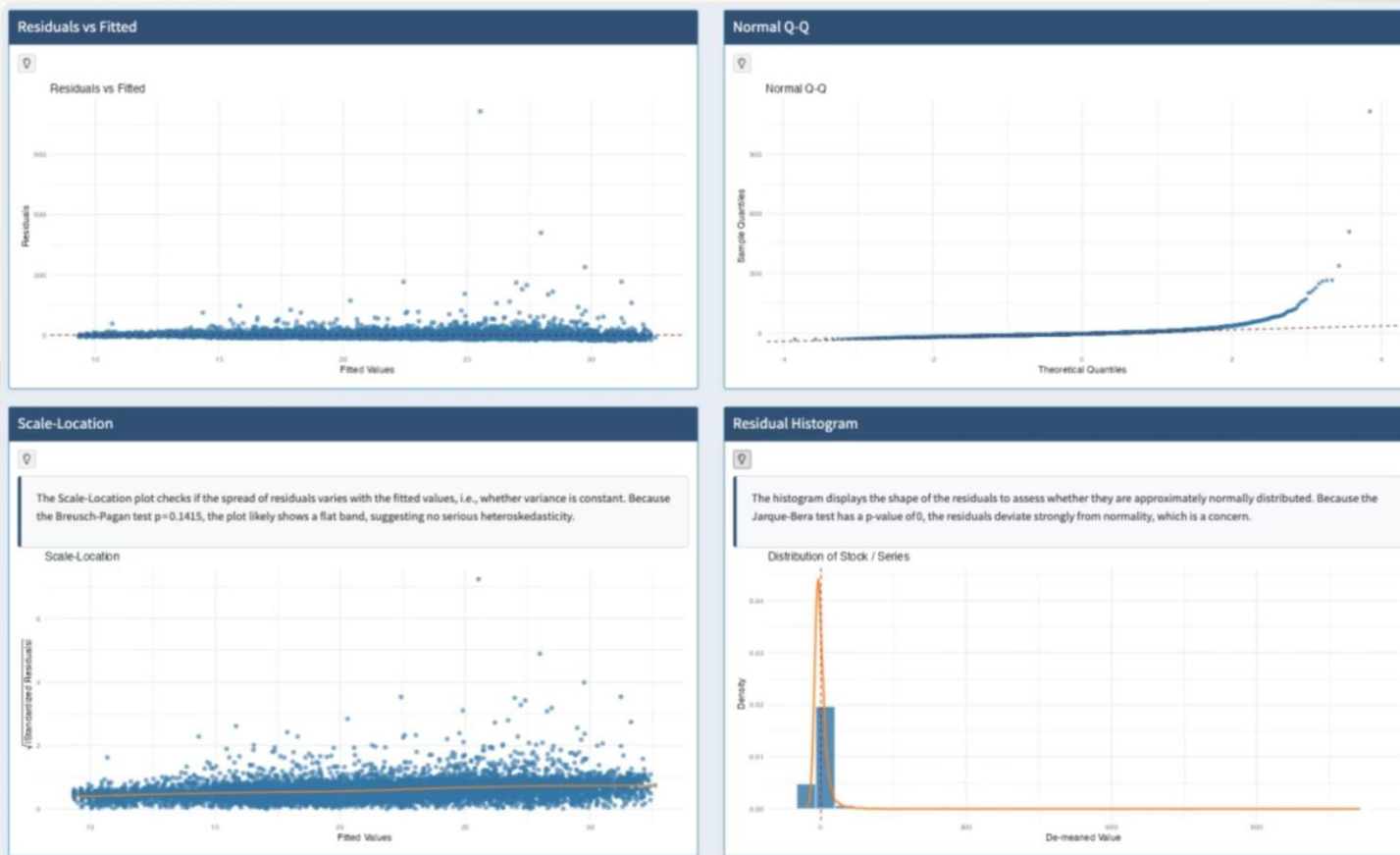
Inspect raw data, structure and statistical properties before modelling

Setup → Diagnostics → Transformation

One interface from data preparation to model refinement.

Users can prepare data and specify the analysis before moving directly into model-aware diagnostics.

From Statistical Tests to Visual Diagnostics



LINEARITY

Residuals vs Fitted · look for patterns

NORMALITY

Q-Q + Histogram · check distribution

CONSTANT VARIANCE

Scale-Location · check residual spread

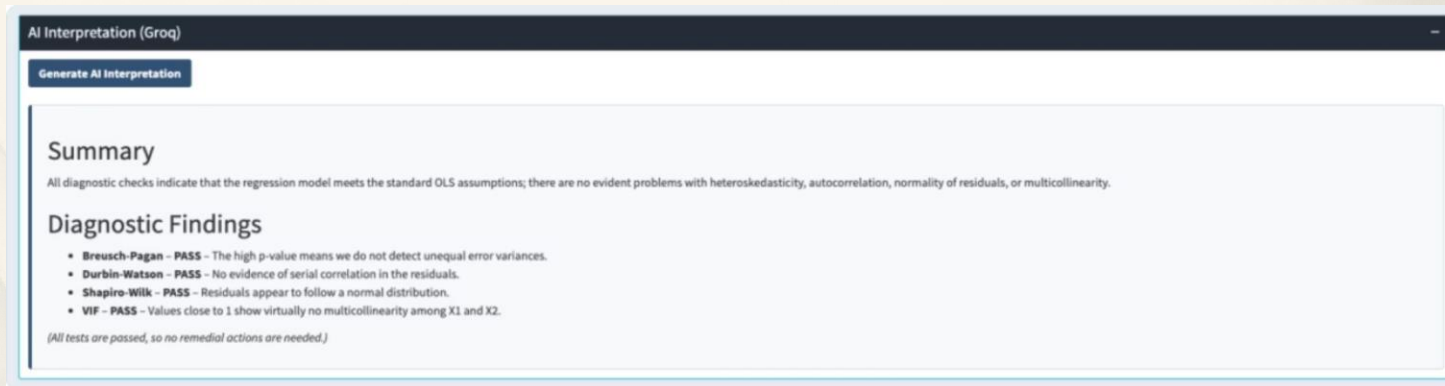
INTERPRETATION

Visual checks complement formal tests

Example shown: BP $p = 0.1415 \rightarrow$ no serious heteroskedasticity · Jarque-Bera $p \approx 0 \rightarrow$ strong departure from normality

Statistics First — AI Interpretation Second

AI translates computed diagnostic results into a concise, plain-language interpretation and suggested actions.



AI explains the diagnostics — it does not calculate or replace them.

01 — COMPUTE

diag_lm runs the statistical diagnostics

02 — SUMMARIZE

Only computed diagnostic results are prepared

03 — INTERPRET

AI explains findings and recommended actions

OPT-IN · NO RAW DATA · WORKS WITHOUT API KEY

User-controlled · Diagnostic summaries only · AI remains optional

Real-Data Case Study: Hourly Wages

Model: $\log(\text{hourly wage}) \sim \text{experience} + \text{age} + \text{female}$ · N = 832

+0.95% / year

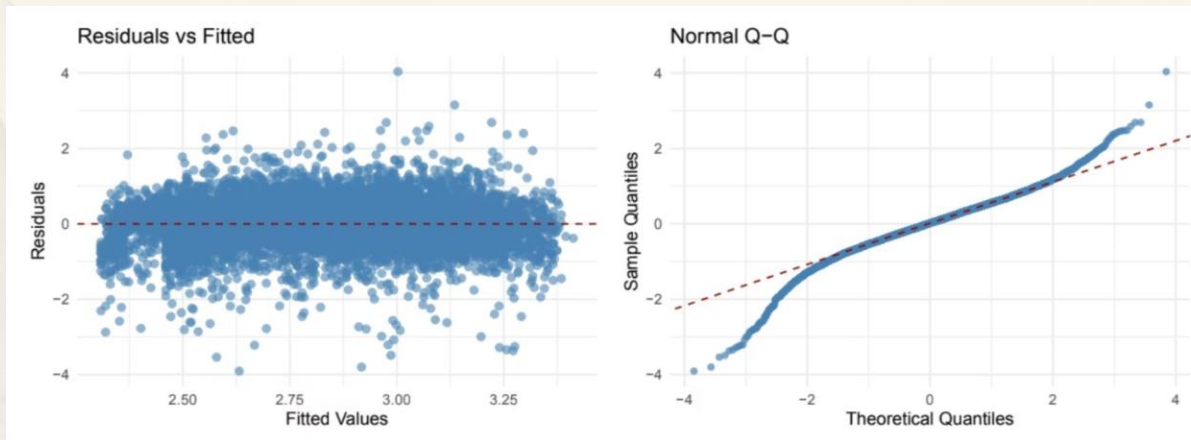
Experience

+1.09% / year

Age

-14.05%

Female indicator



Residual spread changes across fitted values → consistent with heteroskedasticity

Q-Q tails deviate strongly from reference line → non-normal residuals

Diagnostic Results

[!] Breusch-Pagan — CHECK · Heteroskedasticity

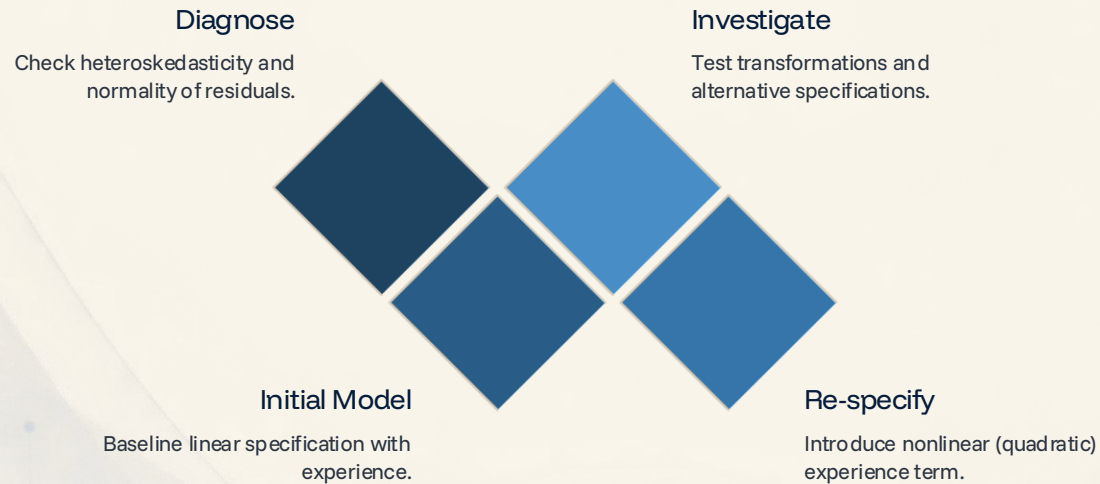
[!] Shapiro-Wilk — CHECK · Non-normal residuals

[OK] Durbin-Watson — PASS

[OK] VIF 4.13 — PASS

The coefficients are interpretable, but diagnostics reveal remaining model problems that motivate re-specification.

Diagnostics Guide Model Improvement



Experience β

+0.0530 — positive wage growth

Experience² β

-0.0010 — diminishing marginal return

The quadratic term captures a **concave earnings profile**: wages rise with experience, but the marginal gain decreases over time — consistent with standard human capital theory.

"Diagnostics are not simply PASS/FAIL checks — they guide model refinement."

What Did We Achieve?

Model-Aware

One framework for linear, logistic, time-series, and panel diagnostics.

Validated

Controlled simulations, real data, and automated software testing.

Interactive

Shiny interface with optional AI-assisted interpretation.

1

FIT

2

DIAGNOSE

3

UNDERSTAND

4

IMPROVE

"diag_lm turns separate diagnostic procedures into a reusable, interactive workflow."

Thank You

Questions & Discussion

We'd love to hear your thoughts, feedback, and questions